

(20)

Example 5.

- A diesel engine power plant has one 700 kW and two 500 kW generating units. The fuel consumption is 0.28 kg per kWh and the calorific value of fuel oil is 10200 kcal/kg. Estimate:
- the fuel oil required for a month of 30 days and
 - overall efficiency.
- Plant capacity factor = 40%

Solution

$$\begin{aligned} \text{(i) Maximum energy that can be produced in a month} &= \text{Plant Capacity} \\ &\quad \times \text{Hours in a month} \\ &= (700 + 2 \times 500) \times (30 \times 24) \\ &= 1700 \times 720 \\ &= 1,224,000 \text{ kWh} \end{aligned}$$

$$\text{Plant Capacity factor} = \frac{\text{Actual energy produced}}{\text{Max. energy that could have been produced}}$$

$$0.4 = \frac{\text{Actual energy produced}}{1,224,000}$$

$$\begin{aligned} \text{Actual energy produced} &= 0.4 \times 1,224,000 \\ &= 489,600 \text{ kWh} \end{aligned}$$

$$\begin{aligned} \text{and fuel oil consumption in a month} &= 0.28 \times 489,600 \\ &= 137,088 \text{ kg} \end{aligned}$$

$$\therefore \text{The fuel oil required for a month of 30 days} = \underline{\underline{137,088 \text{ kg}}}$$

P.T.O

(ii)

$$\begin{aligned}\text{Output} &= 489,600 \text{ kWh} = 489600 \times 860 \text{ kcal} \\ \text{Input} &= 137,088 \times 10,200 \text{ kcal}\end{aligned}$$

$$\therefore \text{Overall efficiency} = \frac{\text{Output}}{\text{Input}} = \frac{489600 \times 860}{137,088 \times 10,200} = 0.301$$
$$= \underline{\underline{30.1\%}}$$

(*) Gas Turbine Power Plant

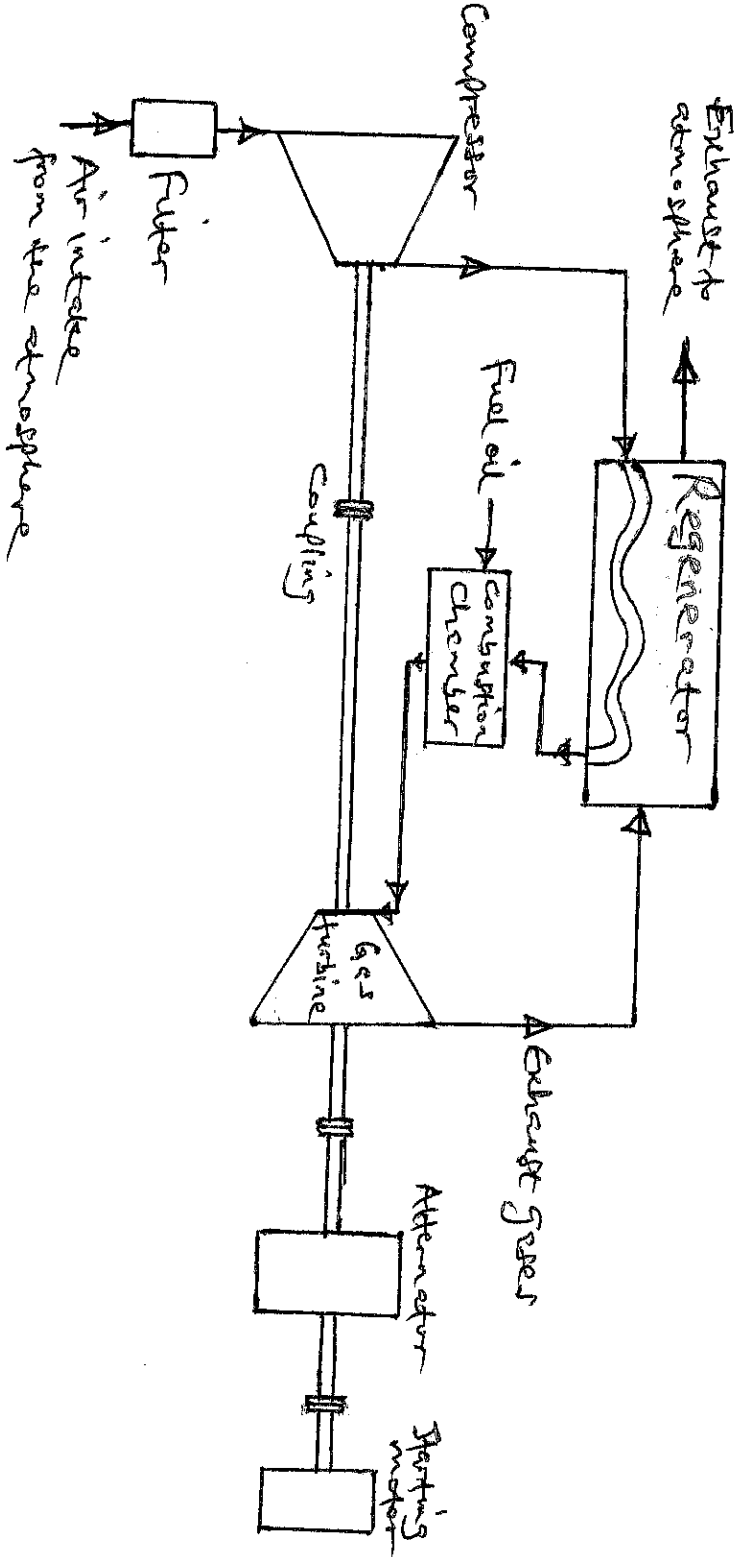
Gas turbine power plant is a power plant that employs the use of gas turbine as the prime mover for the generation of electrical energy.

In a gas turbine power plant, air is used as the working fluid. The air is compressed by the compressor and is led to the combustion chamber where heat is added to the air, thus raising its temperature. Heat is added to the compressed air either by burning fuel in the chamber or by the use of air heaters. The hot and high pressure air from the combustion chamber is then passed to the gas turbine where it expands and does the mechanical work. The gas turbine coupled to the alternator drives the alternator thereby converting the mechanical energy into electrical energy.

Here, the compressor, gas turbine and the alternator are mounted on the same shaft so that a part of mechanical power of the turbine can be utilised for the operation of the compressor.

Gas turbine power plants are being used as standby plants for hydro-electric power plants as a starting plant for drawing auxiliary power in power plants etc.

The schematic diagram of the gas turbine plant is shown on the next page.



Schematic arrangement of a gas turbine power plant.

Components of the Gas turbine Power Plant

The main components of the gas turbine power plant are:

- (i) Compressor,
- (ii) Regenerator,
- (iii) Combustion Chamber,
- (iv) Gas turbine,
- (v) Alternator, and
- (vi) Starting motor.

Compressor:- This is generally of the rotary type. Air at atmospheric pressure is drawn by the compressor via the filter which removes the dust from the air. The rotary blades of the compressor push the air between the stationary blades to raise its pressure. The high pressure ^{air} is available at the output of the compressor.

Regenerator:- This device is used to recover the heat from the exhaust gases of the turbine. The exhaust gases are passed through the regenerator before entering the combustion chamber. The regenerator consists of a nest of tubes contained in a shell. The compressed air from the compressor passes through the tubes in a shell. The compressed air from the combustion chamber enters the tubes on its way to the combustion chamber. In this way, compressed air is heated by the hot exhaust gases.

(iii) Combustion Chamber :- The air at high pressure from the compressor to the combustion chamber via the regenerator receives heat at the combustion chamber from the burning oil. The oil is injected through the burner into the chamber at high pressure to ensure atomisation of oil and its thorough mixing with air. The mixture now attains a very high temperature and the combustion gases are then delivered to the gas turbine.

(iv) Gas turbine :- The product of combustion (mixture of gases at high temperature and pressure) are received at the gas turbine and are passed over the turbine blades in order to expand and do the mechanical work.

(v) Alternator :- The alternator coupled to the gas turbine converts the mechanical energy of the turbine into electrical energy. The output from the alternator is given to the bus-bars through transformer, circuit breaker etc.

(vi) Starting motor :- For the turbine to start, the compressor has to be started. For this purpose, an electric motor is mounted on the same shaft as that of the turbine. The motor is energised by the batteries. Once the unit starts, a part of mechanical power of the turbine drives the compressor and no more need of the motor at that moment.

Selection of Site for Gas Turbine Power Plants

The following factors should be considered while selecting a site for gas turbine power plant.

1. Distance from load centre :- The site should be as near to the load centre as possible so that transmission costs and losses are reduced.
2. Availability of land :- The ^{land} should be available and at a cheap rate and should be of high bearing capacity to withstand the loads of the plant and vibration transmitted to the foundations from compressors and turbines.
3. Availability of fuel :- It should be easily available and at a reasonable rate.
4. Transportation facilities :- The transmission facilities should be available.
5. Distance from populated area :- The site should be away from thick populated areas because of noise and exhaust.

Fuels for Gas Turbines

A variety of fuels (Solid, liquid and gaseous) are available for use in gas turbines. Coal and peat are not used mainly because of coal handling and ash handling problems.

Petroleum fuel (Kerosene, gas oil, diesel oil etc) is limited in their usage because of higher fuel cost. Nowadays, the gas turbine power plants use mainly natural gas and liquid petroleum fuels. Natural gas contains 80% methane and small fraction of other gases.

Advantages of a Gas Power Plant

1. It is simple in design as compared to steam power plant since no boilers and their auxiliaries are required.
2. It is smaller in size as compared to steam power plant of the same capacity.
3. Initial and operating costs are much lower than that of equivalent steam power plant.
4. It requires less water as no condenser is used.
5. Maintenance charges are quite small.
6. It can be started quickly even from cold conditions.
7. There are no standby losses. In steam power plant, these losses occur because boiler is kept in operation even when the steam turbine is supplying no load.

Disadvantages of a Gas Power Plant

1. Net power output is low because a greater part of power developed by the turbine is used in driving the compressor.
2. The overall efficiency of such plant is low (about 20%) because the exhaust gases from the turbine contain significant heat.
3. The temperature of combustion chamber is quite high (3000° F) because that its life is comparatively reduced.
4. Noisy operation etc.

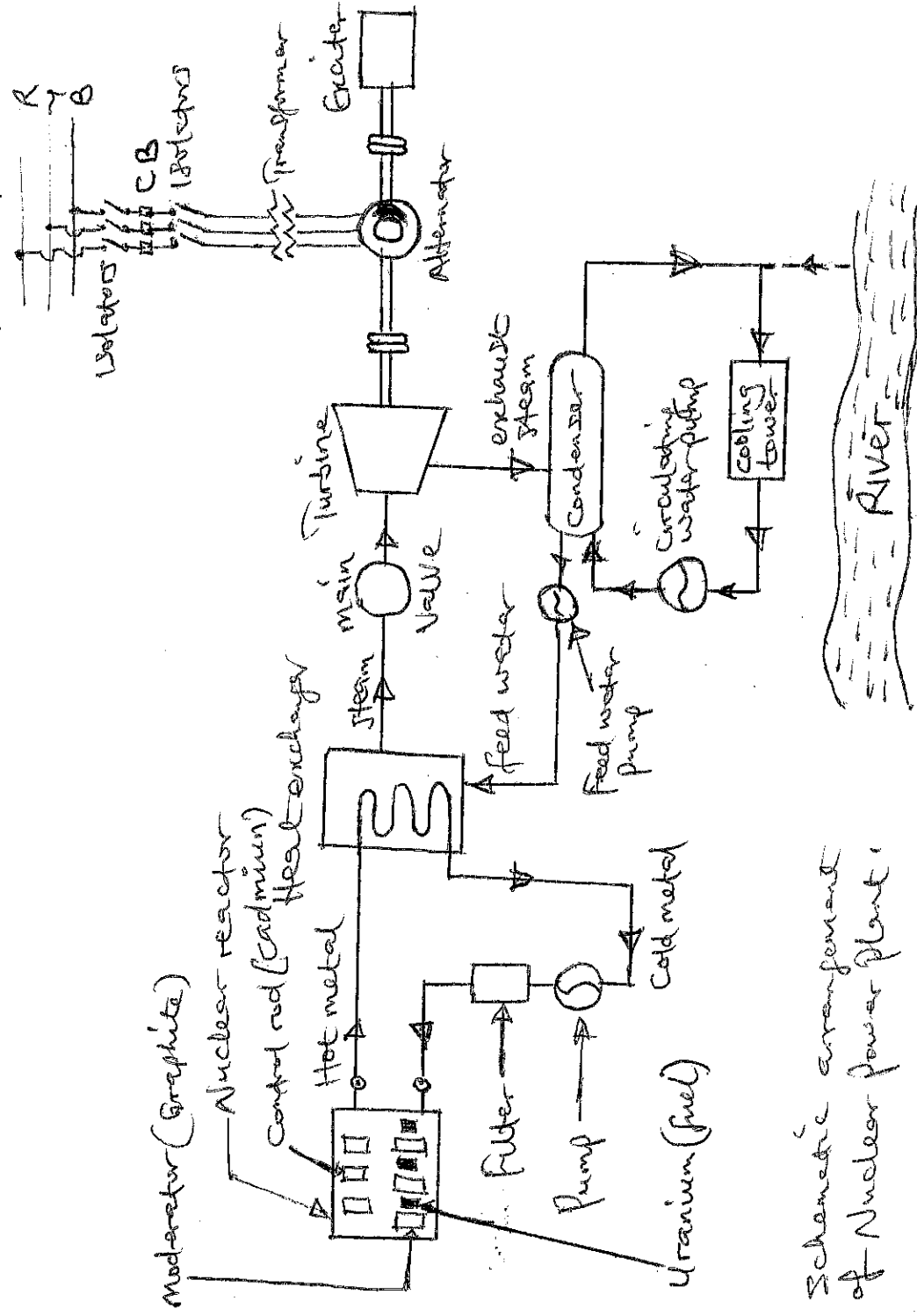
Nuclear Power Plant

A nuclear power plant is the type of power plant in which nuclear energy is converted into electrical energy.

In nuclear power plant, heavy element such as Uranium (^{235}U) or Thorium (^{232}Th) are subjected to nuclear fission (breaking up nuclei of heavy atoms into nearly two equal parts with the release of huge amount of energy) in a special apparatus known as a reactor. The heat energy thus released is utilised in raising steam at high temperature and pressure. The steam runs the steam turbine which converts steam energy into mechanical energy. The turbine drives the alternator which converts the mechanical energy into electrical energy.

Huge amount of electrical energy can be produced from a relatively small amount of nuclear fuel as compared to other conventional types of power plants. Complete fission of 1 kg of Uranium can produce as much energy as can be produced by burning of 4,500 tons of high grade coal. World reserves of these fuels are considerably higher than those of conventional fuels (coal, oil and gas).

The schematic representation of the nuclear power plant is shown in the figure below:



Schematic arrangement of Nuclear power plant.

Elements of the Nuclear Power Plant.

The component arrangement of the Nuclear Power Plant can be divided into the following main stages:

- (i) Nuclear reactor,
- (ii) Heat exchanger,
- (iii) Steam turbine,
- (iv) Alternator.

(i) Nuclear Reactor :- It is an apparatus in which nuclear fuel (U^{235}) is subjected to nuclear fission. It controls the chain reaction (continued nuclear fission) so as to prevent explosion due to the fast increase in the energy released.

It is a cylindrical steel pressure vessel and houses the fuel rods of uranium, moderator (graphite) and control rods (Cadmium). The fuel rods constitute the fission material and release huge amount of energy when bombarded with slow moving neutrons. The Moderator consists of graphite rods which enclose the fuel rods. The moderator slows down the neutrons before they bombard the fuel rods. The moderator slows of Cadmium and are inserted into the reactor. Control rods are strong neutron absorber and thus regulates the supply of neutrons for fission. When the control rods are pushed in deep enough, they absorb most of the fission neutrons and hence few are available for chain reaction - instead then stops. When they are withdrawn, more fission occur with more chain reaction and an increased heat. However raising the of the control rods is done automatically based on the needed - need. The heat produced in the reactor is removed by the coolant - generally a Sodium metal. The coolant carries the heat to the heat exchanger.

(ii) Heat exchanger :- The coolant gives up the heat to the heat exchanger which is utilised in raising the steam. After giving up the heat, the coolant is again fed to the reactor.

(iii) Steam turbine :- The steam produced in the heat exchanger is fed to the steam turbine through a valve. After doing a useful work in the turbine, the steam is exhausted to the condenser. The Condenser

Set Condenses the Steam which is fed to the heat exchanger through the feed water pump.

(IV) Alternator: - The Steam turbine drives the alternator which converts mechanical energy into electrical energy. The output of the alternator is delivered to the bus bars through the transformer, circuit breaker and isolators.

Selection of Site for Nuclear Power Plant

The following factors should be kept in mind when selecting site for a nuclear power plant.

1. Availability of water: - As sufficient water is required for cooling purposes, the plant site should be located where sufficient quantity of water is available e.g. near a river or sea-side.

2. Availability of space for the disposal of waste: - Space and arrangement for disposal of radioactive waste should be made to avoid health hazards. Should be buried in deep trench or disposed off in sea away from the sea shore.

3. Distance from populated areas: - The site should be selected away from the populated areas as there is a danger of presence of radioactive substance in the atmosphere near the plant.

4. Transportation facilities: - This is required in order to transport the heavy equipment during erection / construction and facilitate movement of staff employed in the plant.

5. Type of land: - The land should be strong enough to support the heavy reactors.

6. Nearness to load centre: - Should be situated near the load centre in consistent with safety considerations in order to reduce transmission costs.

Advantages of the Nuclear Power Plant

1. The amount of fuel required is quite small. Therefore, there is a considerable saving in the cost of transportation, storage etc.
2. It requires less space as compared to other type of the same size.

3. It has low operational charges as small amount of fuel is used for producing bulk electrical energy.
4. It can be located near the load centres, therefore, the cost of primary distribution is reduced.
5. There is ^{assurance of} continuous supply of electrical energy for thousands of years since there is large deposits of nuclear fuels available all over the world.
6. It ensures reliability of operation.

Disadvantages of the Nuclear Power Plant

1. The fuel used is expensive and difficult to recover.
2. The capital cost is very high as compared to other types of plants.
3. The erection and commissioning of the plant requires greater technical know-how.
4. The fission by-products are generally radioactive and may cause a dangerous amount of radioactive pollution.
5. Maintenance charges are high. Salary bill of the maintenance staff is also high as specially trained personnel are required to handle the plant.
6. Nuclear power plants are not well suited for varying loads since the reactor does not respond to the fluctuations of load efficiently. Cooling water requirements are very heavy (more than twice the water required for the same size coal-fired steam power plant). Hence cooling towers required are larger and costlier than those of conventional steam power plant.

Assignment 3 (Submit next week)

Describe the following types of nuclear reactors:

- ① Pressurised water reactor (PWR)
- ② Boiling water reactor (BWR)
- ③ Advanced gas-cooled reactor (AGCR)
- ④ Light water graphite-moderated reactor (LWR)
- ⑤ Fast neutron reactor (FNR)
- ⑥ Liquid Metal Cooled Reactor (LMCR)
- ⑦ Fast Breeder Reactor (FBR)

Power Reactors

Power reactors commonly employed in nuclear power plants are:

1. Boiling Water Reactor (BWR): — It is the simplest type of water reactor. It has a steel pressure vessel surrounded by a concrete shield. Fuel used is enriched uranium oxide. Ordinary water is used both as moderator and coolant. The steam is generated in the reactor itself. Feed water enters the reactor vessel at the bottom and takes the heat produced due to fission of fuel and gets converted into steam. This steam leaves the reactor at the top and gets converted into steam through the turbine and condenser, returns to the reactor. Uranium fuel elements are arranged in a particular lattice form inside the pressure vessel containing water.

2. Pressurized Water Reactor (PWR)

Using enriched uranium oxide, clad in zircalloy as fuel. It has fuel rods arranged vertically in the core. The pressure vessel is steel. Water under pressure is used both as coolant and moderator. The pressure vessel and the heat exchanger are surrounded by a concrete shield. In this reactor, bulk boiling water is prevented as the water is pressurized to about 150 atmosphere. The hot water from the reactor flows to a heat exchanger.

(Steam generator) where its heat is transferred to the feed water to generate steam. The secondary cooling operates at low pressure. The primary coolant then flows from the heat exchanger to the primary circulating pump which pumps it back to the reactor.

The steam is condensed in the condenser and the condensate returns to the heat exchanger forming a closed circuit. The primary circuit of the PWR contains a pressurizer (pressure vessel with an electric heating coil at the bottom and a water spray at the top).

3. Gas Cooled Reactor (GCR): — This type of reactor employs a gas (CO_2 or helium) in place of water as the coolant and graphite as the moderator. A heat exchanger is necessarily required.

Gas is circulated through the reactor core and the heat exchanger by means of a blower or a gas compressor. A large portion of gas is required, due to poor heat transfer qualities for circulation resulting in increased power consumption for auxiliaries. Graphitic or moderator is less effective than water and would require a large volume of core in such reactor, the heat removed by gas cooling will be better. The tubes in the heat exchanger through which water is circulated should have fins on their surface to improve the rate of heat transfer.

Power Plant Safety

Power plants are complex environments that house high-voltage equipment, flammable materials, and hazardous chemicals.

A momentary lapse in safety protocols or equipment failure can have catastrophic consequences, resulting in injuries, loss of life, and significant infrastructure damage.

Plant safety and industrial accident prevention measures aim first at preventing accidents, at the plants where hazardous substances are used - and in the event of an accident, minimize the consequent health and environmental effects.

Power Plant Safety Tips

The followings identify safety precautions individuals should follow when working in or near a boiler or other equipment found at power plants.

1. When viewing flames or furnace conditions, always wear tinted goggles or tinted shield to protect the eyes from harmful light intensity and flying ash or slag particles.
2. Do not stand directly in front of open ports or doors especially when they being opened.
3. Do not enter a confined space until it has been cooled, purged of combustible and dangerous gases and properly ventilated.

4. Never open or enter a rotating equipment until it has come to a complete stop and its circuit breaker is open and any other drive devices are immobilized.
5. Do not inspect for tube leak locations until metal and refractory surfaces are cool, and ash accumulations are removed.
6. Be prepared for hot water in drums and headers when removing man hole plates and hand hole covers.
7. Wear goggles and respirators when cleaning the fire side of the boiler, breeching or chimney.
8. Store all oily rags or waste in approved containers to prevent fires caused by spontaneous combustion.
9. Use hand Shields when visually inspecting the furnace fire.
10. When handling any type of rod in the furnace, especially in coal-fired furnaces, be prepared for falling slag striking the rod.
11. Never enter a vessel, especially a boiler drum, until all steam and water valves, including drain and blowdown valves, have been closed or locked.
12. Never use flammable or volatile fluids in confined spaces.
13. Always secure the drive mechanism of dampers, gates and doors before passing through them.

NB All other Safety guidelines regarding Confined Space entry should be known and duly followed.

Assignment 4 (Submit next week)

List thirty (30) Safety policies to be observed in Power Plants.

